

Trackify

The AI-Powered Vintage Ledger

Contributors:

- Mario Alexandru Pastor
- Remus Gabriel Bercut
- Andrei Bogdan Dumitrescu



TRACKIFY LEDGER

AUTHENTICATION

Email Address

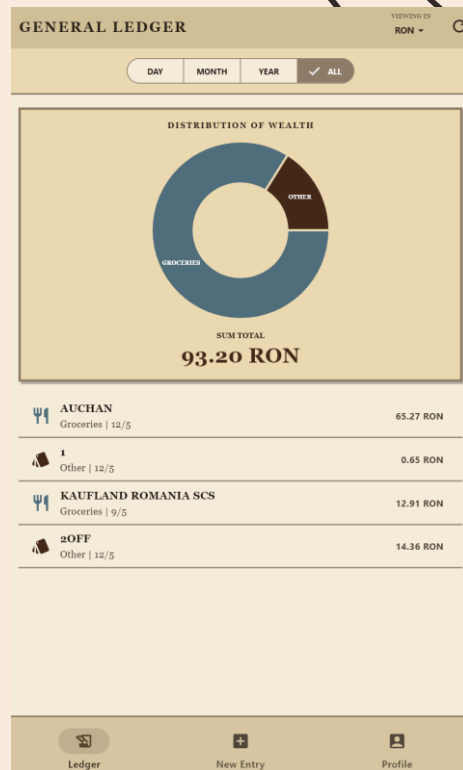
Passphrase

ACCESS LEDGER SESSION

NEW SIGNATORY? REGISTER HERE

Why Trackify?

- **The Problem:** Financial tracking is tedious, and privacy is often surrendered to big-tech cloud providers.
- **The Solution:** A local first based ledger that feels like a physical book but thinks like an AI.
- **Core Value:** Total privacy through self-hosted Opaque Tokens and AI automation.



Under the Hood

Frontend

Flutter
Mobile & Windows Desktop

Backend

FastAPI (Python)
Neon PostgreSQL

Google Gemini API
as the AI Engine for photo
processing

Database

Dual-layer storage:

- SQLite for local
- Neon for Cloud

Roles & Task Distribution

- **Member 1 (The Architect):**
Backend API, Neon DB integration, Gemini AI prompt engineering and WebSocket communication layer.
- **Member 2 (The AI Specialist):**
Offline Sync logic, offline database integration and resync with Neon when active
- **Member 3 (The UI/UX Lead):**
Flutter frontend, Vintage theme implementation & sound, Opaque Token Auth

AI Vision & WebSockets

1. **The Engine:** Google Gemini AI extracts Store, Amount, and Category from photos.
2. **Real-time Web-Sockets:** The app doesn't just wait. The server broadcasts a *RECEIPT_PROCESSED* signal and a pop-up on the screen when the AI is done.
3. **Automation:** Reduces manual entry time by 90%.

NEW LEDGER ENTRY

TRANSACTION DETAILS

 Establishment Name

\$

 Amount Transacted

 Classification

Other

 Transaction Date

2026-05-12

RECORD IN PERMANENT LEDGER

 SCAN RECEIPT

 FROM GALLERY

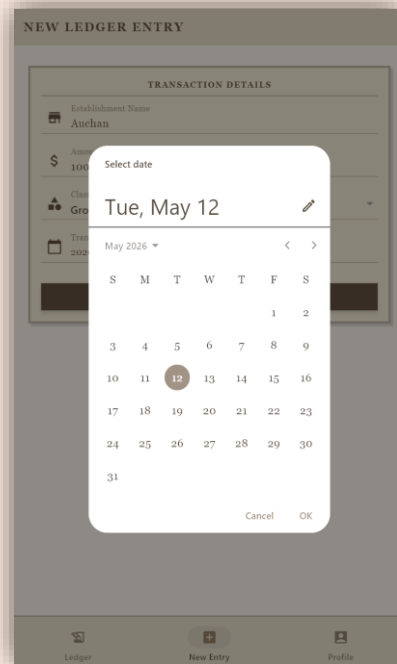
Ledger

New Entry

Profile

The Offline Synchronization Loop

- **Local Persistence:** Uses SQLite to record entries even in Airplane Mode.
- **Connectivity Listener:** Automatically detects when the user is back online. Re-login necessary to save entries into online database.
- **Background Sync:** Pushes pending *is_synced = 0* rows to the cloud without interrupting the user.



Visual & Auditory Identity

Haptic Sounds

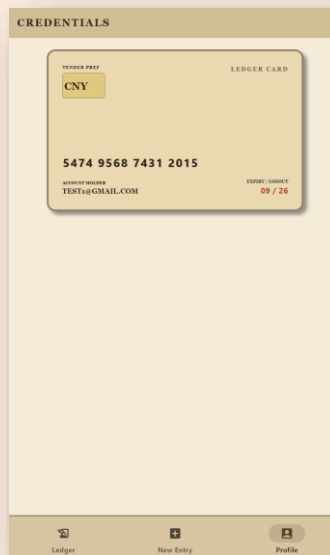
Custom auditory feedback when changing currencies



Aesthetic

Vintage Ledger theme

- Cream paper
- Ink-dark text
- Georgia fonts



UX Focus

Minimalist navigation using an *IndexedStack* for zero-latency screen switching.

Project Status & Future

Current State: Fully functional Offline-First prototype with AI integration.

Future Roadmap:

- Add functionality to scan receipt directly with the smartphone's camera
- Sync the offline database with the online one immediately when internet is found (not necessary to re-login)
- Multi-user shared ledgers, add receipts in different currencies at the same time



Thank You

Trackify: Personal Finance, Architecturally Secure.